

6.7 PreQuiz: Polynomials — Markscheme

1. [Maximum mark: 7]

Let $f(x) = x^2 + 2x - 15$.

(a) Write $f(x)$ in factored form $(x - a)(x - b)$. [2]

$$f(x) = (x - 3)(x - (-5)) = (x - 3)(x + 5) \quad \text{(M1)(A1)}$$

Accept $(x + 5)(x - 3)$.

(b) Write down the x -intercepts. [1]

$$x = 3 \text{ and } x = -5 \quad \text{(A1)}$$

(c) Write down the y -intercept. [1]

$$f(0) = -15, \text{ so the } y\text{-intercept is } (0, -15). \quad \text{(A1)}$$

(d) Write $f(x)$ in vertex form $(x - h)^2 + k$. [1]

$$f(x) = (x + 1)^2 - 16 \quad \text{(A1)}$$

Or equivalently, $(x - (-1))^2 + (-16)$, i.e. $h = -1$, $k = -16$.

(e) Find the coordinates of the vertex. [2]

$$\text{Axis of symmetry: } x = \frac{-5 + 3}{2} = -1 \quad \text{(A1)}$$

$$f(-1) = 1 - 2 - 15 = -16$$

$$\text{Vertex: } (-1, -16) \quad \text{(A1)}$$

2. [Maximum mark: 7]

Consider the function $g(x) = -(x - 4)^2 + 9$.

(a) Write down the coordinates of the vertex. [1]

$$\text{Vertex: } (4, 9) \quad \text{(A1)}$$

(b) Write down the equation of the axis of symmetry. [1]

$$x = 4 \quad \text{(A1)}$$

(c) Show that $g(x) = -x^2 + 8x - 7$. [2]

$$g(x) = -(x - 4)^2 + 9 = -(x^2 - 8x + 16) + 9 \quad \text{(M1)}$$

$$= -x^2 + 8x - 16 + 9 = -x^2 + 8x - 7 \quad \text{(A1)(AG)}$$

(d) Solve $g(x) = 0$ to find the x -intercepts. [3]

$$-x^2 + 8x - 7 = 0 \quad \textbf{(M1)}$$

$$x^2 - 8x + 7 = 0$$

$$(x - 1)(x - 7) = 0 \quad \textbf{(A1)}$$

$$x = 1 \text{ or } x = 7 \quad \textbf{(A1)}$$

3. [Maximum mark: 7]

The line $y = 2x$ and the parabola $y = x^2 - 3$.

(a) Show that the x -coordinates of intersection satisfy $x^2 - 2x - 3 = 0$. [2]

At intersection: $x^2 - 3 = 2x$ (M1)

$x^2 - 2x - 3 = 0$ (A1)(AG)

(b) Show that $x^2 - 2x - 3 = 0$ has two distinct real roots. [1]

$\Delta = (-2)^2 - 4(1)(-3) = 4 + 12 = 16 > 0$ (A1)

Since $\Delta > 0$, there are two distinct real roots.

(c) Factorise $x^2 - 2x - 3$. [1]

$(x - 3)(x + 1)$ (A1)

(d) Find the x -coordinates of the points of intersection. [1]

$x = 3$ and $x = -1$ (A1)

(e) Find the coordinates of both points of intersection. [2]

$x = 3: y = 2(3) = 6 \implies (3, 6)$ (A1)

$x = -1: y = 2(-1) = -2 \implies (-1, -2)$ (A1)

4. [Maximum mark: 7]

The equation $2x^2 - kx + 8 = 0$ has two equal roots.

(a) Write down the discriminant in terms of k . [1]

$\Delta = (-k)^2 - 4(2)(8) = k^2 - 64$ (A1)

(b) Find the values of k . [2]

Equal roots: $\Delta = 0$

$k^2 - 64 = 0 \implies k^2 = 64$ (M1)

$k = \pm 8$ (A1)

(c) For $k = 8$, find the repeated root. [2]

$2x^2 - 8x + 8 = 0 \implies x^2 - 4x + 4 = 0$ (M1)

$(x - 2)^2 = 0 \implies x = 2$ (A1)

(d) State the values of k for which the equation has no real roots. [2]

$$\Delta < 0 \implies k^2 - 64 < 0 \implies k^2 < 64 \quad \textbf{(M1)}$$

$$-8 < k < 8 \quad \textbf{(A1)}$$